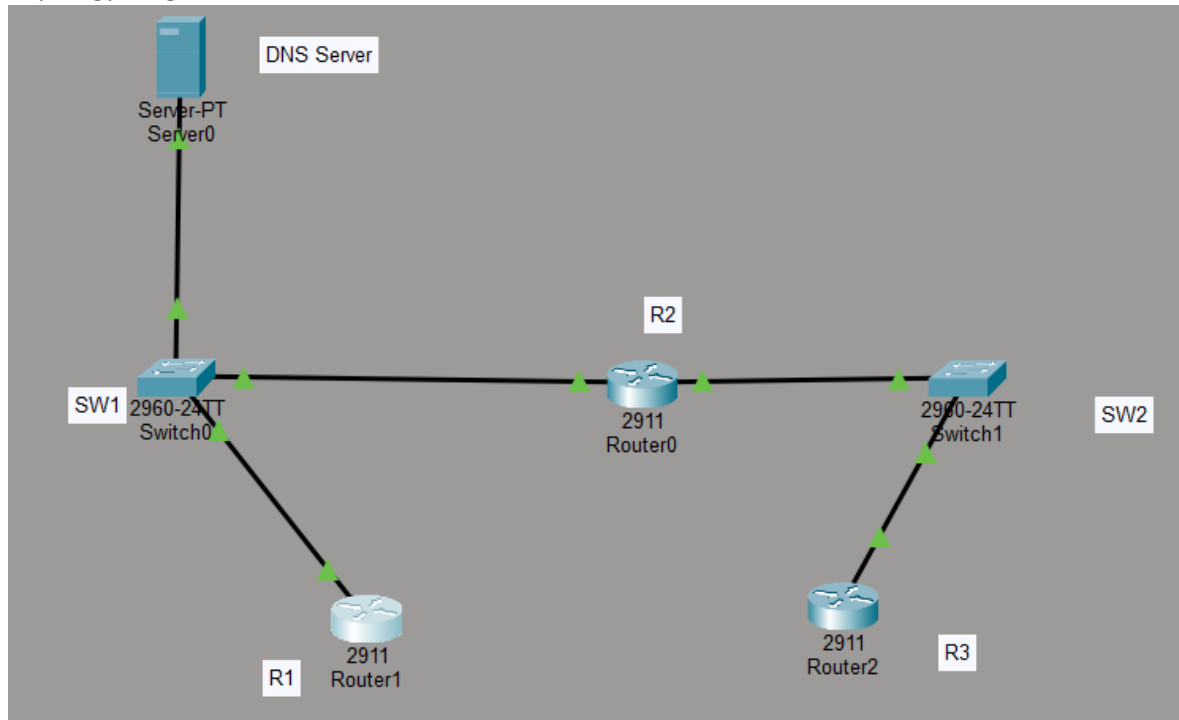
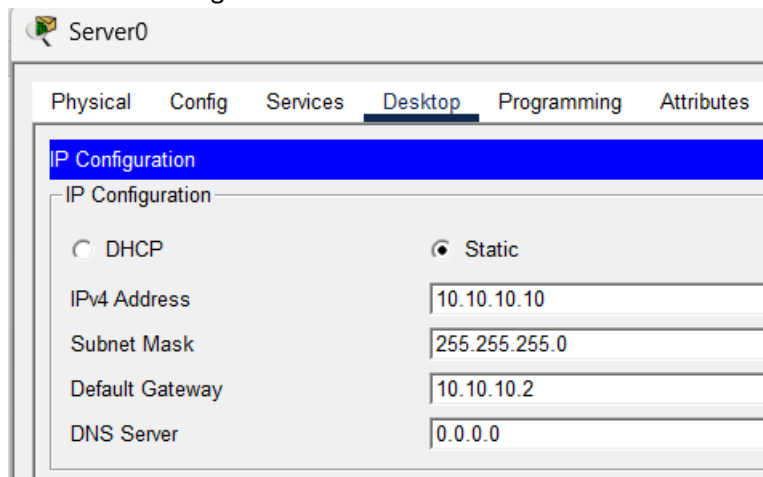


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20260924
NET512 Week 03 Practical's

➤ Topology Diagram:



➤ DNS Server Configurations:



Services Desktop Programming Attributes

DNS

DNS Service ☒ On ☐ Off

Resource Records

Name Type A Record

Address

Add
Save
Remove

No.	Name	Type	Detail
0	r1	A Record	10.10.10.1
1	r2	A Record	10.10.10.2
2	r3	A Record	10.10.20.1

➤ Router Configurations:

Physical Config CLI Attributes

IOS Command Line Interface

```

Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R2
R2(config)#int fa0/0
%Invalid interface type and number
R2(config)#int g0/0
R2(config-if)#ip address 10.10.10.2 255.255.255.0
R2(config-if)#no shutdown

R2(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

R2(config-if)#description LAN1 to sw1
R2(config-if)#exit
R2(config)#int fa1/0
%Invalid interface type and number
R2(config)#int g1/0
%Invalid interface type and number
R2(config)#int g0/1
R2(config-if)#ip address 10.10.20.2 255.255.255.0
R2(config-if)#no shutdown

R2(config-if)#
  
```

- Testing Name resolution ping success instead of an IP address

```
R2>en
R2#ping r1
Translating "r1"...domain server (10.10.10.10)
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.1, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/3/13 ms

R2#ping r2
Translating "r2"...domain server (10.10.10.10)
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/4/6 ms

R2#ping r3
Translating "r3"...domain server (10.10.10.10)
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.20.1, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms

R2#
```

- Cleared ARP

```
R1>en
R1#clear arp-cache
R1#ping 10.10.10.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms
```

- Show ARP after clearing

```
R1#show arp


| Protocol | Address    | Age (min) | Hardware Addr  | Type | Interface          |
|----------|------------|-----------|----------------|------|--------------------|
| Internet | 10.10.10.1 | -         | 00E0.A31A.8601 | ARPA | GigabitEthernet0/0 |
| Internet | 10.10.10.2 | 0         | 000C.CFD2.4401 | ARPA | GigabitEthernet0/0 |


R1#
```

20. What information does an ARP entry give you that a routing table entry does not?

- A routing table tells you *where* to send a packet next, an ARP entry fills in the information that is missing, like the real MAC address that goes with the next IP hop to where the data packet needs to go.

21. Why does a router need to perform an ARP lookup even after it already knows the correct route from its routing table?

- Because the routing table gives the router a next-hop IP address to where data packet goes to, but the Ethernet doesn't deliver frames using IP addresses, it uses MAC addresses. Routing decides the path and ARP makes the actual delivery on that path possible.